

Introduction to Robotics, Lecture 1

Introduction

Position and Orientation

Change of frames

MIPT

Center for Cognitive Modeling

About Me (1)



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About Me (2)

Education

- *Bachelor* — BMSTU (МГТУ им Н.Э. Баумана) «CAD developer»
- *Master & PHD* — Innopolis University (IU) «Robotics»

Job

- *Lead Engineer, Senior Research Fellow* — MIPT, Center for Cognitive Modeling: SLAM and robotics hardware development
- *Associate Professor* — IU: Supervisor

Link to my channel

Why This Course?

Our laboratory works with AI and has real **robot manipulators** and **mobile robots**.

However, using a real robot requires more than developing or training an algorithm. We need to understand:

- How robot configuration, pose, and motion are described
- What physical and kinematic constraints limit motion
- How sensing, noise, synchronization, and calibration affect the system
- Why singularities, odometry drift, and SLAM failures occur
- Where CV, RL, and other AI methods fit into the robot architecture

This course provides the **robotics foundation** needed to work with these systems competently.

Course Goal

To develop the **system-level understanding**

needed to formulate, analyze, and diagnose problems involving **robot manipulators** and **mobile robots**,

and to apply CV, RL, and other AI methods under the **correct assumptions**

Understand the robotics problem before choosing the algorithm

Course Outline

- **Robot Manipulators (Weeks 1–5):** coordinate frames and transformations, forward and inverse kinematics, Jacobians, singularities, parallel robots, trajectories, and industrial robot commands
- **Mobile Robots (Weeks 7–9):** locomotion mechanisms, differential-drive, Ackermann and omnidirectional models, odometry, control tasks, and navigation
- **Sensors, Calibration, and State Estimation (Weeks 10–11):** encoders, IMUs, LiDARs, cameras and GNSS; noise, synchronization and sensor fusion; intrinsic, extrinsic, temporal and hand–eye calibration

  Simultaneous Localization and Mapping (SLAM) (Weeks 12–13): dead reckoning, landmark-based SLAM, pose graphs, data association, loop closure, classical and learning-based approaches, evaluation metrics, and failure modes

Grading criteria

What will be evaluated on the course

- Qz: Quizzes: 20%
- HWs: Homework assignments: 40%
- HP: Hardware Projects: 30%
- FE: Final Exam: 10%
- Extra: Slide fixes in Github: 5%

Late policy: -50% of max grade for a task. One week late is allowed.

Scale:

- A: 80 — 100%
- B: 50 — 79.99%
- C: 29.99 — 49.99%
- D: 0 — 29.99% or less than 50% by any criterion.

Quizzes

Purpose: Quizzes help us to understand what you didn't get during the previous lecture.

Format: 5 — 15 *min* closed book on a paper sheet, in the beginning of a lecture.

Lab Goals

Solve math problems on a whiteboard and defend you HWs.

Hardware Projects

From models to real robotic systems

The course includes three hardware-centered projects:

1. Industrial Manipulator Task Programming
2. Robot–Camera Calibration and Validation
3. Mobile Robot Identification, Odometry, and SLAM

Each project follows the same engineering workflow:

Model → Configure or calibrate → Execute → Measure → Diagnose

Hardware Projects

Organizational problems

- Three manipulators and three small mobile robots are available
- Students may work in teams of two
- Each team submits one shared report
- Project results are presented and defended as a team
- A personal robotic platform may be used with prior approval

Do not just make the robot work—explain why it works and why it fails.

Project 1: Industrial Manipulator Programming

Objective: Program and execute a representative industrial task using a real robot manipulator.

The project includes:

- Defining the base, tool, and work-object coordinate frames
- Planning a pick-and-place, inspection, or similar task
- Selecting appropriate joint-space and Cartesian-space commands
- Using MoveJ, MoveL, and, if supported, MoveC
- Checking joint limits, singularities, and possible collisions
- Comparing planned and observed robot behavior

Result: A working hardware demonstration and a report explaining the selected commands, assumptions, limitations, and possible failures.

Project 2: Robot --- Camera Calibration

Objective: Solve and validate an eye-in-hand or eye-to-hand calibration problem.

The project includes:

- Defining all coordinate frames and transformation directions
- Collecting measurements from multiple robot poses
- Estimating the unknown camera-to-robot transformation
- Validating the result using previously unseen poses
- Measuring the resulting task-space error
- Analyzing the effects of noise, pose selection, and incorrect frame conventions

Result: A calibrated robot-camera system that can relate visual observations to robot coordinates, together with quantitative validation.

Project 3: Mobile Robot Identification and SLAM

Objective: Identify the motion model of a mobile robot and evaluate its influence on odometry, localization, and mapping.

The project includes:

- Identifying kinematic parameters or the relationship between commands and robot velocity
- Executing and measuring a predefined trajectory
- Quantifying odometry error and accumulated drift
- Running a localization or SLAM pipeline
- Comparing results before and after model identification

Result: A mobile robot demonstration and an evidence-based analysis of the difference between predicted, measured, and estimated motion.

Course Materials and Acknowledgements

The materials are reorganized and supplemented with the instructor's own explanations, examples, and practical assignments and based on the following sources:

- **Robotics 1 and Robotics 2**
  Sapienza University of Rome — Alessandro De Luca
- **Introduction to Robotics**
  Stanford University — Oussama Khatib
- **Advanced Robotics Manipulation**
  Innopolis University — Alexandr Klimchik
- **Autonomous Mobile Robots**
  ETH Zürich — Roland Siegwart, Davide Scaramuzza, and colleagues
- **Modern Robotics: Mechanics, Planning, and Control**
  Northwestern University — Kevin M. Lynch and Frank C. Park

What Is a Robot? (1)





What Is a Robot? (2)









What Is a Robot? (3)

Wikipedia

A robot is an automatic device designed to perform various mechanical operations according to a predefined program.

What Is a Robot? (3)

Wikipedia

A robot is an automatic device designed to perform various mechanical operations according to a predefined program.

GOST R 60.0.0.4-2019

An actuated mechanism programmable in two or more axes, possessing a degree of autonomy and capable of moving within its environment to perform intended tasks.

What Is a Robot? (3)

Wikipedia

A robot is an automatic device designed to perform various mechanical operations according to a predefined program.

GOST R 60.0.0.4-2019

An actuated mechanism programmable in two or more axes, possessing a degree of autonomy and capable of moving within its environment to perform intended tasks.

ISO/TC 299

A programmable actuated mechanism with a degree of autonomy that performs locomotion, manipulation, or positioning.

New standards

International Organization for Standardization



According to the above ISO definition in 2012, these are NOT robots

- software (“bots”, AI, Robotic Process Automation - RPA)
- voice assistants
- ATMs (automatic money teller machines)
- cooking machines, smart washing machines, …

and also

- remote-controlled drones, UAV, UGV, UUV
- autonomous cars

} but in the revised standard ISO8373:2021 these are now classified as (autonomous) robotic devices!

The word robot began as cultural imagination

computerized numerically controlled (CNC) machines

~ 1950

mechanical telemanipulators



robot manipulators

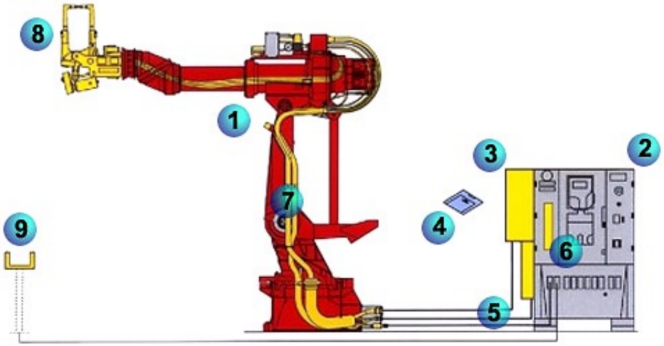
1970
Unimation PUMA



- with respect to the ancestors
  - flexibility of use
  - adaptability to a priori unknown conditions
  - accuracy in positioning
  - repeatability of operation

An industrial robot system is more than an arm

1. Comau SMART H robot
2. C3G Plus controller
3. Welding control box
4. Application software
5. Air/water supply
6. SWIM Board
7. Integrated cables
8. Welding gun
9. Auxiliary devices in the robotic cell (servo-controlled axes)



SWIM = Spot Welding Integrated Module

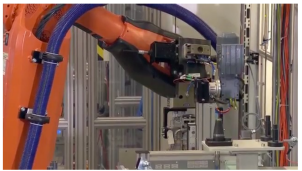
Industrial tasks impose different constraints

- At BMW car production line with ABB robots

video



video



pick-and-place
with end-effector
to reorient part

pick-and-place
with support
to reorient part

What a robot should do and what cannot do

yet

video



spray painting
very unhealthy
for human operators

video



assembly of flexible
or complex parts
(here a car dashboard)

⇒ human-robot **collaboration**
(co-bots or co-workers)

Manipulation, assembly, packaging, coating, welding, machining, inspection, and HRC are source examples, not universal capability evidence.

Mobile Robots in Real-World Applications



Delivery

Starship Technologies



Inspection

ANYmal, ETH Zurich



Transportation

Waymo

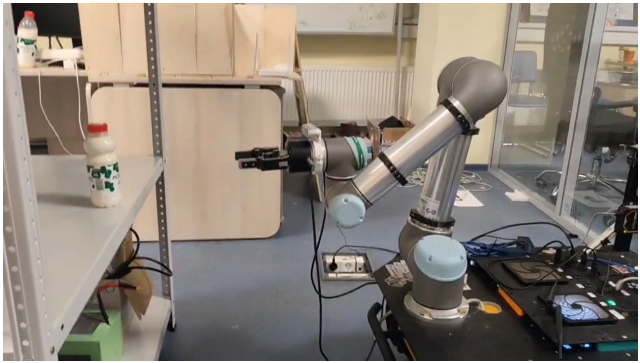


Mobile manipulation

AGIBOT G2

Mobile robot with manipulator





Pretest (1/2)

Robots, motion, and coordinate frames

1. What distinguishes a robot from a conventional automated machine?
2. What are the main components of a robotic system?
3. What is a degree of freedom? How many degrees of freedom does a free rigid body have in 3D space?
4. Do the coordinates of a physical point change when we choose a different coordinate frame?

Pretest (2/2)

Position, orientation, and pose

5. What is the difference between the position and orientation of an object?
6. What methods can be used to represent the orientation of an object in 3D space?
7. How can we represent both the position and orientation of an object relative to a coordinate frame?

Pretest Discussion (1/2)

Core ideas

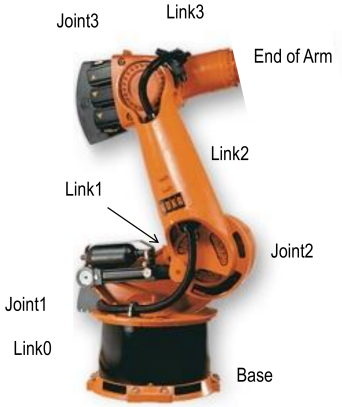
- **Robot**: A system that senses its environment, processes information, and acts on the physical world.
- **Main components**: Sensors, controller or computer, actuators, mechanical structure, and power supply.
- **Degree of freedom**: An independent variable required to describe motion.
- **Free rigid body in 3D**: Six degrees of freedom: three translations and three rotations.

Pretest Discussion (2/2)

Position, orientation, and pose

- **Position**: Location of an object, commonly represented by a vector
$$\mathbf{p} = \begin{bmatrix} x & y & z \end{bmatrix}^T.$$
- **Orientation**: Angular relationship between two coordinate frames. Common representations include Euler or roll–pitch–yaw angles, rotation matrices, axis–angle, and quaternions.
- **Pose**: Position and orientation considered together.
- **Rotation order**: In general,
$$\mathbf{R}_1\mathbf{R}_2 \neq \mathbf{R}_2\mathbf{R}_1.$$

Links and Joints



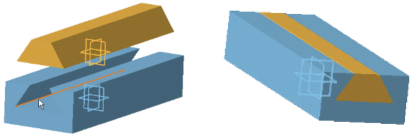
A manipulator consists of **links** and **joints**.

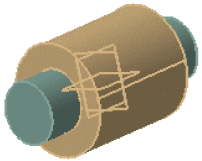
- **Links**: Rigid members between joints.
- **Joints**: Connections that provide relative motion between links.

A robot manipulator has two sections:

- **Body and arm**: Positioning objects in the robot's work volume.
- **Wrist**: Orienting objects.

Prismatic and Revolute Joints





Prismatic Joint (P)

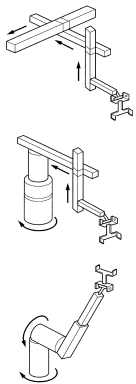
- Linear motion; no rotation.
- Actuation examples: hydraulic or pneumatic cylinder.

Revolute Joint (R)

- Rotary motion.
- Actuation examples: stepper motor or servo motor.

Robot Types (1/2)

Body-and-arm configurations



Cartesian / Rectangular / Gantry (3P)

- Three prismatic joints.
- Linear motion along the *X*, *Y*, and *Z* axes.

Cylindrical (R2P)

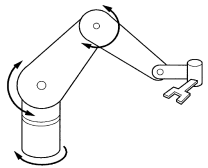
- One revolute joint and two prismatic joints.

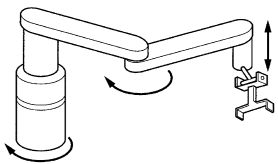
Spherical / Polar (2RP)

- Two revolute joints and one prismatic joint.

Robot Types (2/2)

Body-and-arm configurations





Articulated / Anthropomorphic (3R)

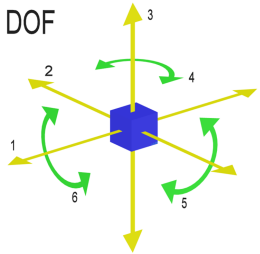
- Three revolute joints.
- Human-arm-like configuration.

SCARA

- Selective Compliance Assembly Robot Arm.
- Illustrated positioning assembly: two revolute joints with parallel axes and one prismatic joint.

How Many Coordinates Describe a Rigid Body?

- **In a plane: 3** — two position coordinates and one orientation angle.
- **In space: 6** — three position coordinates and three orientation parameters.



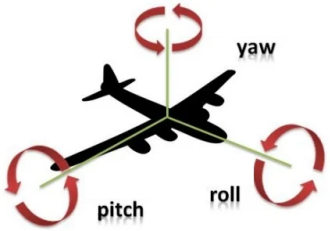
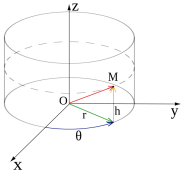


Fig. 8

Position Representations

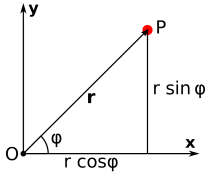
2D

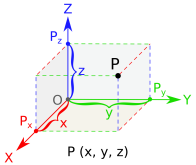
- Cartesian
- Polar

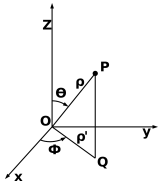


3D

- Cartesian
- Cylindrical
- Spherical







Orientation Representations

Rotation matrix: 9 parameters.

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix}$$

Euler angles: 3 parameters.

$$(\alpha, \beta, \gamma)$$

Axis-angle: 4 parameters.

$$\left(\begin{bmatrix} e_x \\ e_y \\ e_z \end{bmatrix}, \theta \right) = \left(\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, -\frac{\pi}{2} \right)$$

Quaternion: 4 parameters.

$$(x, y, z, w)$$

Rotation Matrices

$R \in SO(n)$, the **special orthogonal group** of $n \times n$ matrices. The columns form an orthonormal basis:

$$R^T R = I, \qquad R^T = R^{-1}, \qquad \det R = 1.$$

Elementary rotations in 3D, with $c = \cos\varphi$, $s = \sin\varphi$:

$$R_x(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c & -s \\ 0 & s & c \end{bmatrix}, \qquad R_y(\varphi) = \begin{bmatrix} c & 0 & s \\ 0 & 1 & 0 \\ -s & 0 & c \end{bmatrix},$$

$$R_z(\varphi) = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Rotation Matrices

A sequence of rotations has an explicitly specified order:

$$R = R_x(\alpha)R_y(\beta)R_z(\gamma).$$

With $c_a = \cos\alpha$, $s_a = \sin\alpha$ (and similarly for b, c):

$$R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_a & -s_a \\ 0 & s_a & c_a \end{bmatrix} \begin{bmatrix} c_b & 0 & s_b \\ 0 & 1 & 0 \\ -s_b & 0 & c_b \end{bmatrix} \begin{bmatrix} c_c & -s_c & 0 \\ s_c & c_c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

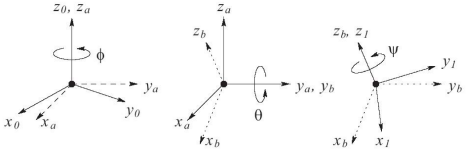
$$= \begin{bmatrix} c_b c_c & -c_b s_c & s_b \\ s_a s_b c_c + c_a s_c & -s_a s_b s_c + c_a c_c & -s_a c_b \\ s_a s_c - c_a s_b c_c & s_a c_c + c_a s_b s_c & c_a c_b \end{bmatrix}.$$

Euler Angles

Rotate about z, then the **new** y, then the **newest** z axis.

$$R_{ZYZ} = R_z(\phi)R_y(\theta)R_z(\psi).$$

$$R_{ZYZ} = \begin{bmatrix} c_\phi c_\theta c_\psi - s_\phi s_\psi & -c_\phi c_\theta s_\psi - s_\phi c_\psi & c_\phi s_\theta \\ s_\phi c_\theta c_\psi + c_\phi s_\psi & -s_\phi c_\theta s_\psi + c_\phi c_\psi & s_\phi s_\theta \\ -s_\theta c_\psi & s_\theta s_\psi & c_\theta \end{bmatrix}$$



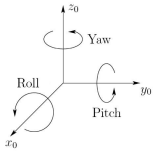
Problem: gimbal lock. $c_\eta = \cos\eta, s_\eta = \sin\eta.$

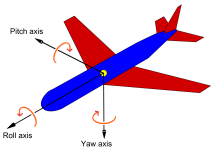
Fixed Angles

Three successive rotations about fixed principal axes: **roll** (x), **pitch** (y), **yaw** (z).

$$R = R_z(\phi)R_y(\theta)R_x(\psi), \qquad c_\eta = \cos\eta,\ s_\eta = \sin\eta.$$

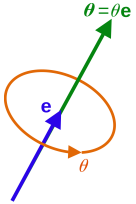
$$R = \begin{bmatrix} c_\phi c_\theta & c_\phi s_\theta s_\psi - s_\phi c_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ s_\phi c_\theta & s_\phi s_\theta s_\psi + c_\phi c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi \\ -s_\theta & c_\theta s_\psi & c_\theta c_\psi \end{bmatrix}$$





Axis-Angle

$$(\text{axis}, \text{angle}) = \left(\begin{bmatrix} e_x \\ e_y \\ e_z \end{bmatrix}, \theta \right) = \left(\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, -\frac{\pi}{2} \right).$$



Unit axis $\mathbf{u} = (u_x, u_y, u_z)^T$;

$c = \cos\theta$, $s = \sin\theta$, $v = 1 - c$.

$$R = \begin{bmatrix} c + u_x^2 v & u_x u_y v - u_z s & u_x u_z v + u_y s \\ u_y u_x v + u_z s & c + u_y^2 v & u_y u_z v - u_x s \\ u_z u_x v - u_y s & u_z u_y v + u_x s & c + u_z^2 v \end{bmatrix}.$$

Quaternions

Quaternions provide a convenient mathematical representation of orientations and rotations in space.

Compared with Euler angles, they make rotation composition convenient and avoid gimbal lock.

$$q = q_r + q_i\mathbf{i} + q_j\mathbf{j} + q_k\mathbf{k}, \qquad s = \|q\|^{-2}.$$

$$R = \begin{bmatrix} 1 - 2s(q_j^2 + q_k^2) & 2s(q_iq_j - q_kq_r) & 2s(q_iq_k + q_jq_r) \\ 2s(q_iq_j + q_kq_r) & 1 - 2s(q_i^2 + q_k^2) & 2s(q_jq_k - q_iq_r) \\ 2s(q_iq_k - q_jq_r) & 2s(q_jq_k + q_iq_r) & 1 - 2s(q_i^2 + q_j^2) \end{bmatrix}.$$

The four components parameterize q; for a unit quaternion, $s = 1$.

Orientation Representations

Task 1

A coordinate frame rotates by 20° about its *x* axis, then by 45° about its **new** *z* axis, and finally by 70° about its **new** *x* axis.

Express the resulting orientation as:

1. A rotation matrix.
2. Euler angles.
3. Fixed angles.
4. A quaternion.
5. An axis–angle pair.

Orientation Representations

Task 1

Answer

Moving-axis rotations give the intrinsic *XZX* sequence:

$$R = R_x(20^\circ)R_z(45^\circ)R_x(70^\circ).$$

$$R \approx \begin{bmatrix} 0.707107 & -0.241845 & 0.664463 \\ 0.664463 & -0.094134 & -0.741369 \\ 0.241845 & 0.965738 & 0.094134 \end{bmatrix}.$$

- **Euler angles (intrinsic *XZX*):** $(20^\circ, 45^\circ, 70^\circ)$.
- **Fixed angles (extrinsic *xzx*):** $(70^\circ, 45^\circ, 20^\circ)$.

Orientation Representations

Task 1

Answer

Quaternion, with the scalar component first:

$$q = q_x(20°)q_z(45°)q_x(70°)$$

$$q \approx 0.653281 + 0.653281\mathbf{i} + 0.161729\mathbf{j} + 0.346829\mathbf{k}.$$

$$\theta = 2\arccos(q_r) \approx 98.421058°, \qquad \mathbf{u} = \frac{(q_i, q_j, q_k)^T}{\sin(\theta/2)}.$$

$$(\mathbf{u}, \theta) \approx \left(\begin{bmatrix} 0.862856 \\ 0.213612 \\ 0.458093 \end{bmatrix}, 98.421058° \right).$$

Summary

1. A rigid-body pose requires **six coordinates in space** and **three in a plane**.
2. We considered five position representations across 2D and 3D.
3. We examined four orientation representations and how they relate to one another.

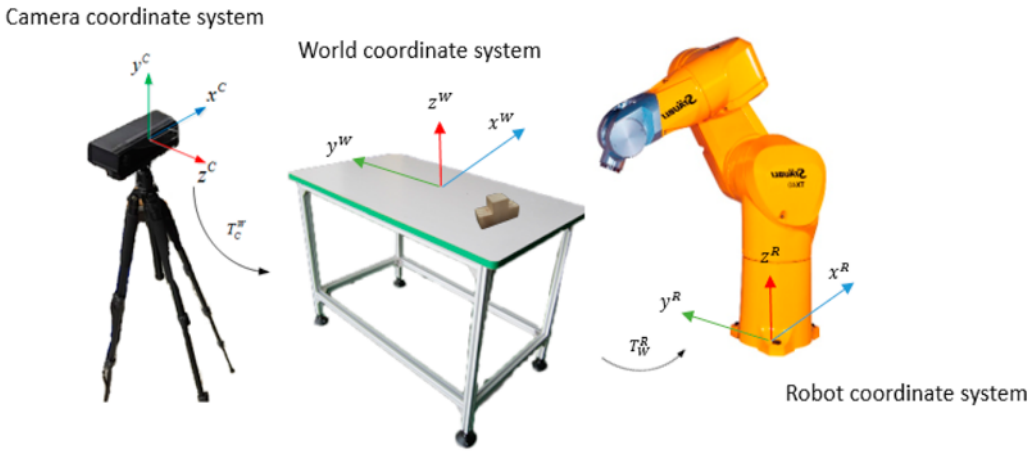
Changing Basis

Case Study: Shoe Polishing Robot



Changing Basis

Case Study: Shoe Polishing Robot, How it works



Changing Basis

Two Bros in the World of Changing Basis

$$E' = EA \text{ and } Ex = E'x'.$$

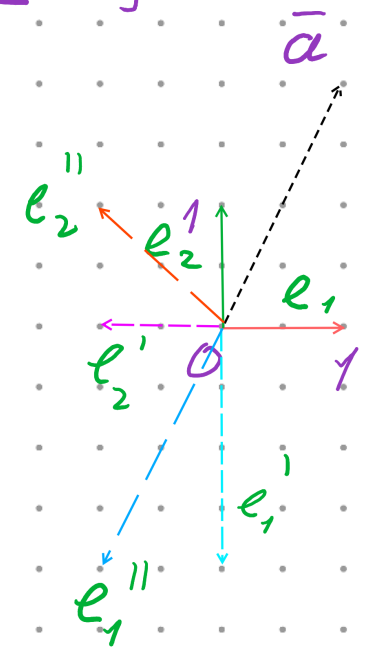
Extended Bro Equation: $Ex = Eb + E'x'$, where

$$E = [e_1\ e_2\ e_3] = \begin{bmatrix} e_{1x} \\ e_{1y} \\ e_{1z} \end{bmatrix}, \begin{bmatrix} e_{2x} \\ e_{2y} \\ e_{2z} \end{bmatrix}, \begin{bmatrix} e_{3x} \\ e_{3y} \\ e_{3z} \end{bmatrix}, E' = [e'_1\ e'_2\ e'_3]$$

Answer from figure

$a = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $a'' = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

$a' = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$



$E' = EA$ $Ex = E'x'$

$x = Ax'$

$E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $E \to E'$

1) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \end{bmatrix} = E' \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

e_2'

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -2 \end{bmatrix} = E' \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

e_1'

$I \begin{bmatrix} 0 & -1 \\ -2 & 0 \end{bmatrix} = E' \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$A_{E \to E'}$

$E' \to E''$; $E' = \begin{bmatrix} 0 & -1 \\ -2 & 0 \end{bmatrix}$

2) $\begin{bmatrix} 0 & -1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = E'' \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

e_1''

$\begin{bmatrix} 0 & -1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} -0.5 \\ 1 \end{bmatrix} = E'' \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

e_2''

$\begin{bmatrix} 0 & -1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 1 & -0.5 \\ 1 & 1 \end{bmatrix} = E'' = \begin{bmatrix} -1 & -1 \\ -2 & 1 \end{bmatrix}$

$A_{E' \to E''}$

$\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -2 & 0 \end{bmatrix} a'$

$\begin{bmatrix} 0 & -0.5 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$

a'

$\begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & -0.5 \\ 1 & 1 \end{bmatrix} a'' \Rightarrow \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

a''

$\begin{bmatrix} 2/3 & 1/3 \\ -2/3 & 2/3 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} = a''$

Changing Basis

Task 2

If vectors **a** and **b** form a basis (you should check it), it is needed to find coordinates **c** in the basis.

$$\mathbf{a} = \begin{bmatrix} -5 \\ -1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \mathbf{c} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$

Changing Basis

Task 2

Answer

$$E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \; E' = \begin{bmatrix} -5 & -1 \\ -1 & 3 \end{bmatrix}$$

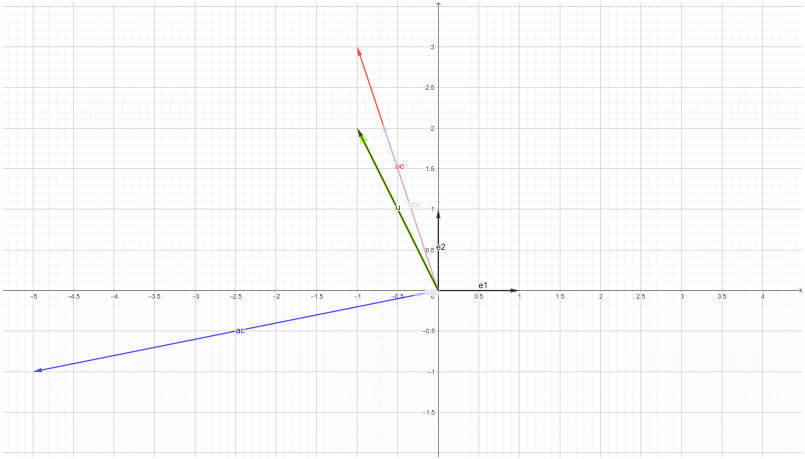
Because we are interested in coordinates, 2nd bro

$$Ec_{old} = E'x \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -5 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} c_{new_x} \\ c_{new_y} \end{bmatrix}$$

$$\begin{bmatrix} c_{new_x} \\ c_{new_y} \end{bmatrix} = (E')^{-1} c_{old} = \begin{bmatrix} \frac{1}{16} \\ \frac{11}{16} \end{bmatrix}$$

Changing Basis

Task 2



Changing Basis

Homogeneous representation

$$\mathbf{V}^{XY} = \begin{bmatrix} \mathbf{V}^X \\ \mathbf{V}^Y \end{bmatrix} = \begin{bmatrix} \mathbf{P}_x \\ \mathbf{P}_y \end{bmatrix} + \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \mathbf{V}^N \\ \mathbf{V}^O \end{bmatrix}$$

What we found by doing a translation and a rotation

$$= \begin{bmatrix} \mathbf{V}^X \\ \mathbf{V}^Y \\ \mathbf{1} \end{bmatrix} = \begin{bmatrix} \mathbf{P}_x \\ \mathbf{P}_y \\ \mathbf{1} \end{bmatrix} + \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{V}^N \\ \mathbf{V}^O \\ \mathbf{1} \end{bmatrix}$$

Padding with 0's and 1's

$$= \begin{bmatrix} \mathbf{V}^X \\ \mathbf{V}^Y \\ \mathbf{1} \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & \mathbf{P}_x \\ \sin\theta & \cos\theta & \mathbf{P}_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{V}^N \\ \mathbf{V}^O \\ \mathbf{1} \end{bmatrix}$$

Simplifying into a matrix form

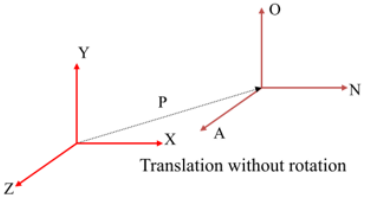
$$\mathbf{H} = \begin{bmatrix} \cos\theta & -\sin\theta & \mathbf{P}_x \\ \sin\theta & \cos\theta & \mathbf{P}_y \\ 0 & 0 & 1 \end{bmatrix}$$

Homogenous Matrix for a Translation in XY plane, followed by a Rotation around the z-axis

Changing Basis

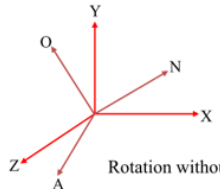
Special Cases of Homogeneous matrices in 3D

H is a 4x4 matrix that can describe a translation, rotation, or both in one matrix



Translation without rotation

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & P_x \\ 0 & 1 & 0 & P_y \\ 0 & 0 & 1 & P_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Rotation without translation

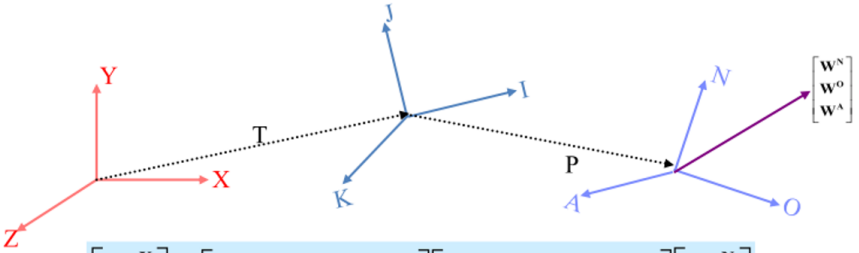
$$\mathbf{H} = \begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation part:

Could be rotation around z-axis x-axis, y-axis or a combination of

Changing Basis

Change the coordinate frame: Case Study



$$\begin{bmatrix} W^X \\ W^Y \\ W^Z \\ 1 \end{bmatrix} = \begin{bmatrix} i_x & j_x & k_x & T_x \\ i_y & j_y & k_y & T_y \\ i_z & j_z & k_z & T_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_i & o_i & a_i & P_i \\ n_j & o_j & a_j & P_j \\ n_k & o_k & a_k & P_k \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} W^N \\ W^O \\ W^A \\ 1 \end{bmatrix}$$

Changing Basis

Task 3

Let us consider two coordinate systems in the plane: O, $\mathbf{e}_1$, $\mathbf{e}_2$ and O', $\mathbf{e}'_1$, $\mathbf{e}'_2$. Point O' has coordinates $(7; -2)$ in the old coordinate system, and vectors $\mathbf{e}'_1$, $\mathbf{e}'_2$ can be obtained from vectors $\mathbf{e}_1$, $\mathbf{e}_2$ by rotating them 60° (a) clockwise; (b) counterclockwise. Find the old coordinates of a point x, y given its new coordinates x', y'.

Changing Basis

Task 3

Answer

Classical way, using Extended 2nd Bro eqn

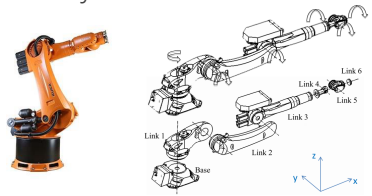
$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ -2 \end{bmatrix} + \begin{bmatrix} \cos 60 & -\sin 60 \\ \sin 60 & \cos 60 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

Or using Homogeneous matrix

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} \cos 60 & -\sin 60 & 7 \\ \sin 60 & \cos 60 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

Example: KUKA Robot Manipulator

From the base frame to the tool frame



$$\mathbf{T} = \mathbf{T}_{\text{Base}}\, T_z(d_0)\, \mathbf{A}\, \mathbf{W}\, T_x(d_6)\, \mathbf{T}_{\text{Tool}},$$

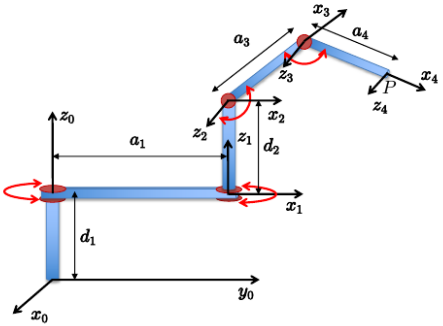
$$\mathbf{A} = R_z(q_1)T_x(d_1)R_y(q_2)T_x(d_2) \cdot R_y(q_3)T_x(d_3)T_z(d_4),$$

$$\mathbf{W} = R_x(q_4)R_y(q_5)R_x(q_6).$$

A: robot manipulator; **W**: robot wrist.

Changing Basis

Task 4



Find the homogeneous transformation matrix ${}^{0}\mathbf{T}_{4}$ for the robot shown.

Given:

- Link dimensions: d_1, a_1, d_2, a_3, a_4.
- Joint angles: θ_1, θ_2, θ_3, θ_4.

The matrix maps coordinates from frame 4 to frame 0:

$$ {}^{0}\bar{\mathbf{p}} = {}^{0}\mathbf{T}_{4}\, {}^{4}\bar{\mathbf{p}}. $$

Changing Basis

Task 4

Answer

Following the frames from the base to the end of the arm:

$$
\begin{aligned}
{}^{0}\mathbf{T}_1 &= T_z(d_1)R_z(90°)R_z(\theta_1)T_x(a_1),\\
{}^{1}\mathbf{T}_2 &= R_z(\theta_2)T_z(d_2)R_x(90°),\\
{}^{2}\mathbf{T}_3 &= R_z(\theta_3)T_x(a_3),\\
{}^{3}\mathbf{T}_4 &= R_z(\theta_4)T_x(a_4).
\end{aligned}
$$

$$
\boxed{{}^{0}\mathbf{T}_4 = {}^{0}\mathbf{T}_1\,{}^{1}\mathbf{T}_2\,{}^{2}\mathbf{T}_3\,{}^{3}\mathbf{T}_4}
$$

T_x, T_z: translations; R_x, R_z: rotations.

Changing Basis

Task 4

Answer

Let $s_i = \sin\theta_i$, $c_i = \cos\theta_i$, $s_{ij} = \sin(\theta_i + \theta_j)$, $c_{ij} = \cos(\theta_i + \theta_j)$.

$$
{}^0\mathbf{T}_4 = \begin{bmatrix} -s_{12}c_{34} & s_{12}s_{34} & c_{12} & p_x \\ c_{12}c_{34} & -c_{12}s_{34} & s_{12} & p_y \\ s_{34} & c_{34} & 0 & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

$$
\begin{aligned}
p_x &= -a_1 s_1 - s_{12}(a_3 c_3 + a_4 c_{34}), \\
p_y &= a_1 c_1 + c_{12}(a_3 c_3 + a_4 c_{34}), \\
p_z &= d_1 + d_2 + a_3 s_3 + a_4 s_{34}.
\end{aligned}
$$

Reference Materials

- A. De Luca — 01 IndustrialRobots
- A. Klimchik — L1 Introduction
- Skillbox — Prik linal 7
- Skillbox — Mod2 latex text
- Skillbox — Mod2
- Skillbox — Prik linal 8

Deserve “A” grade!

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